

WASP-12b

R-0546

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_181021 · created 2026-07-22T05:45:29+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458411.89442 +/- 0.00053 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.095 +/- 0.011
Transit depth (Rp/Rs)^2	0.9 +/- 0.21 [%]
Orbital Inclination (inc)	83.43 +/- 2.29
Airmass coefficient 1 (a1)	1.1398 +/- 0.0098
Airmass coefficient 2 (a2)	-0.1188 +/- 0.0034
Scatter in the residuals of the lightcurve fit is	0.85 %
Best Comparison Star	None
Optimal Aperture	4.66
Optimal Annulus	14.26
Transit Duration (day)	0.1207 +/- 0.0053

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.095 ± 0.011 vs 0.1178 (deviation 1.5σ)
Duration fit vs published	0.1207 d vs 0.125 d (deviation 0.59σ)
Mid-time O–C	-1570.6 min
Depth SNR	6.2
Residual scatter	0.85 %

Light curve

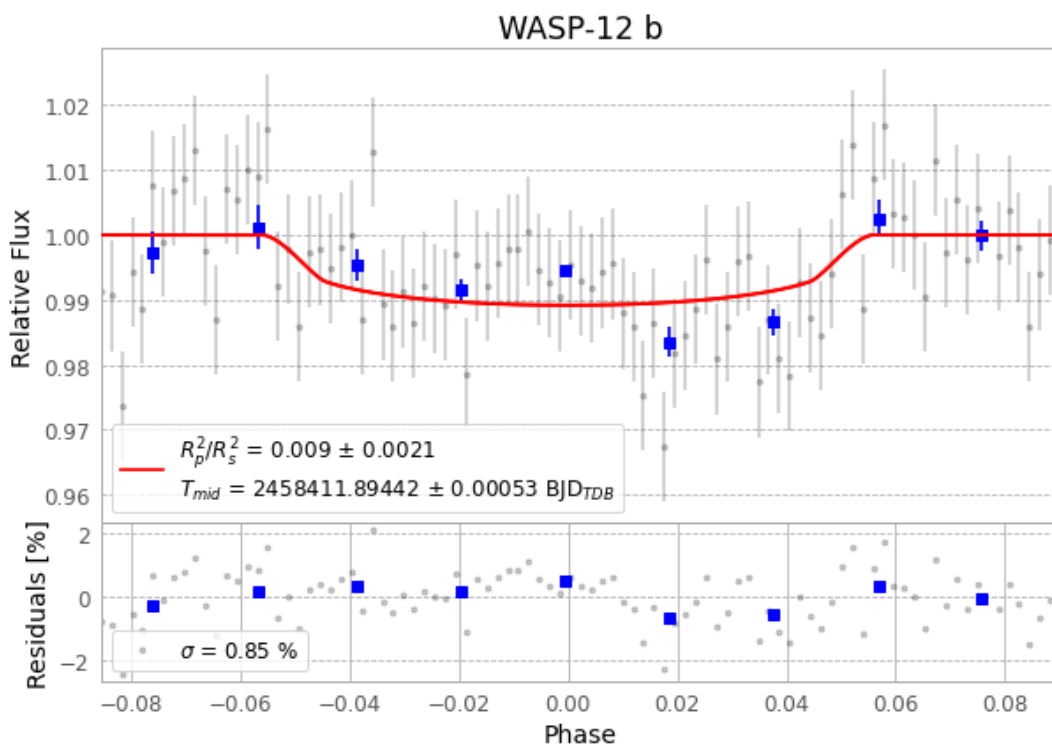


Figure 1 — FinalLightCurve_WASP-12 b_2018-10-20.png (SHA-256 47ed4e8886937aeb...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [319, 155], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6ca03122c57db270...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

93 FITS frames (61 MB), WASP-12181020070912.FITS → WASP-12181020114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-181021.log (147305924d39...), FinalLightCurve_WASP-12 b_2018-10-20.png (47ed4e888693...), FinalLightCurve_WASP-12 b_2018-10-20.csv (3bcd1f5f3505...)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-22 05:45Z.